

Edge Geography is XNLP-hard for Pathwidth and in XP for Tree-Partition Width

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Edge Geography

One of the most famous PSPACE-complete problems.

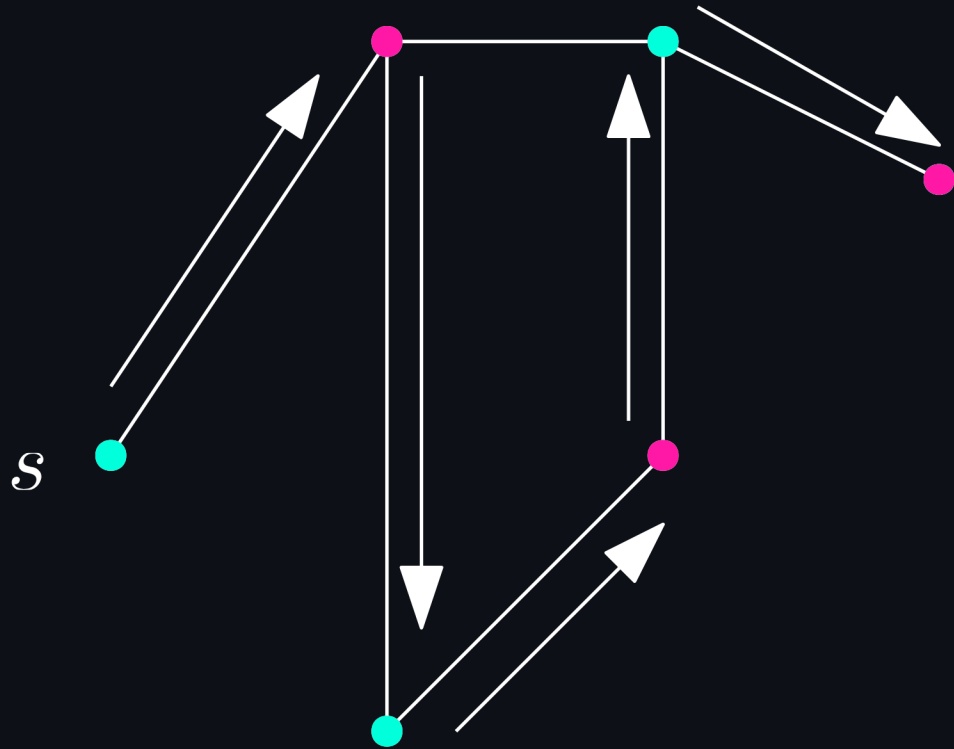
A token starts at a vertex s .

Players alternate:

1. choose an unused edge from the current vertex,
2. move the token along it,
3. delete that edge.

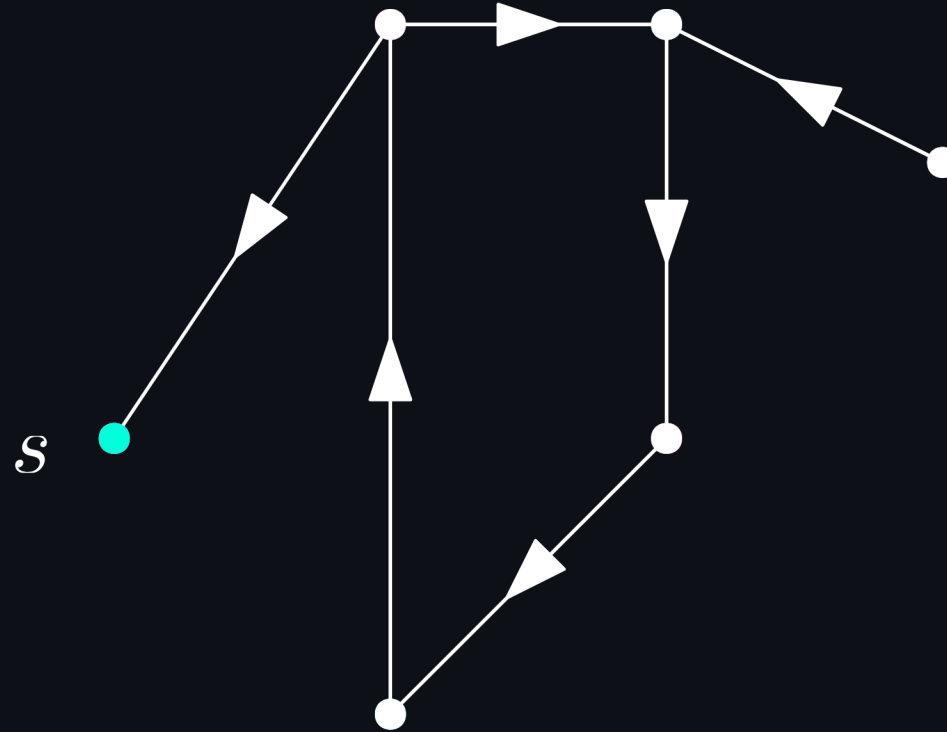
The first player unable to move loses.

Undirected Edge Geography



Player 1 wins!

Directed Edge Geography



Player 2 wins!

Parameterizing the problem

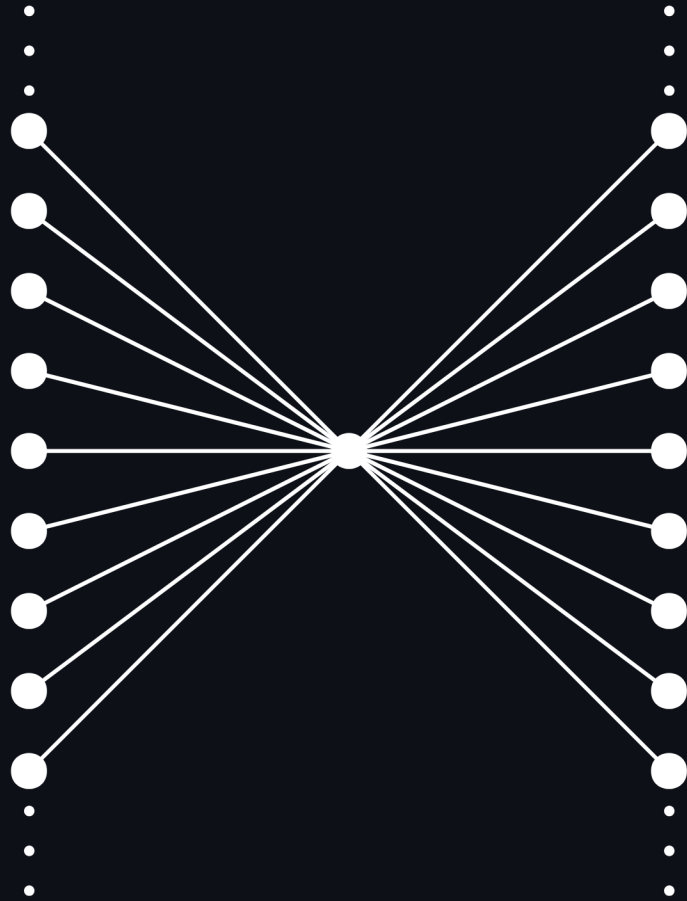
Theorem (Bodlaender, modern language)

(Undirected) Edge Geography is solvable in time $f(k, d) \cdot O(n)$ where k is the treewidth of the graph and d is the maximum degree of any vertex in the graph.

Exponential time becomes "linear time" when two parameters are controlled!

What if we allow d to be unbounded?

Arbitrarily many crossings



when $\Delta(G)$ is unbounded!

Our first result

Theorem

Directed Edge Geography is XNLP-hard parameterized by pathwidth.

XNLP "rules out" fixed-parameter tractability.

Bounded pathwidth $\Rightarrow$ bounded treewidth.

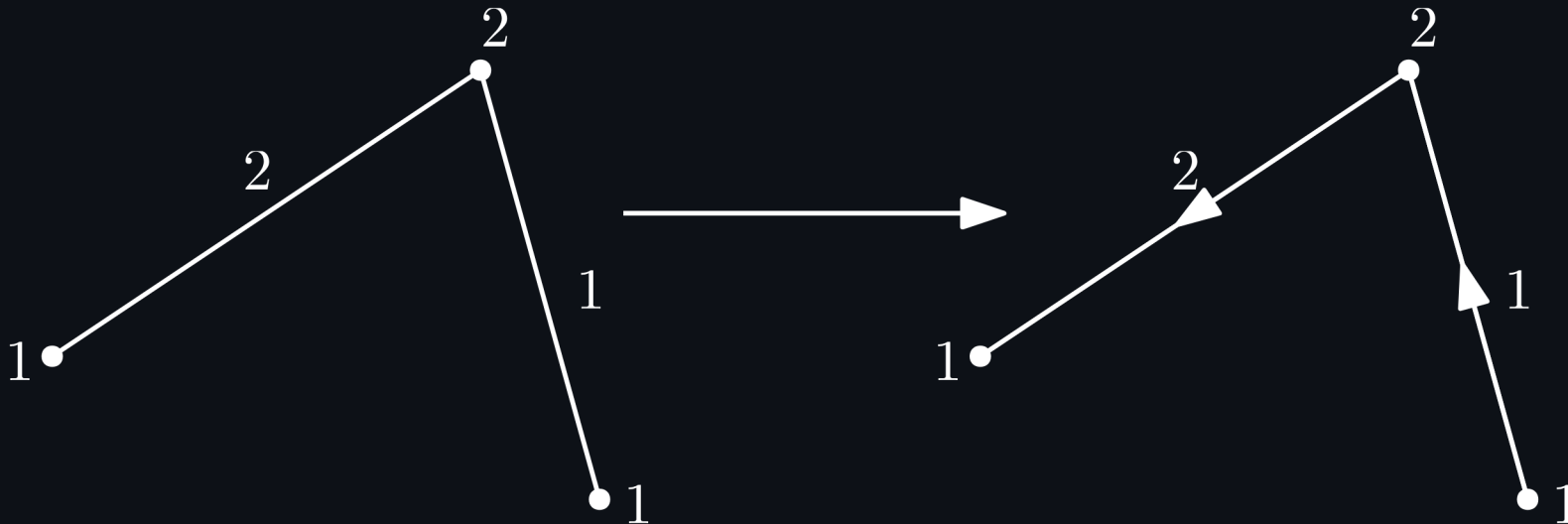
Pathwidth, and thus treewidth, is not enough!

This result transfers to Undirected Edge Geography in a simple way!

Source problem: Chosen Maximum Outdegree

Does there exist an orientation of every edge so that each vertex respects its budget.

$$\sum_{e \in \theta_w(v)} w(e) \leq t(v) \quad \forall v$$



Known XNLP-complete when parameterized by pathwidth.

Reduction idea in two phases

Phase I — Player 1 encodes an orientation

For each input edge $e = \{x, y\}$, Player 1 chooses one of two sides.

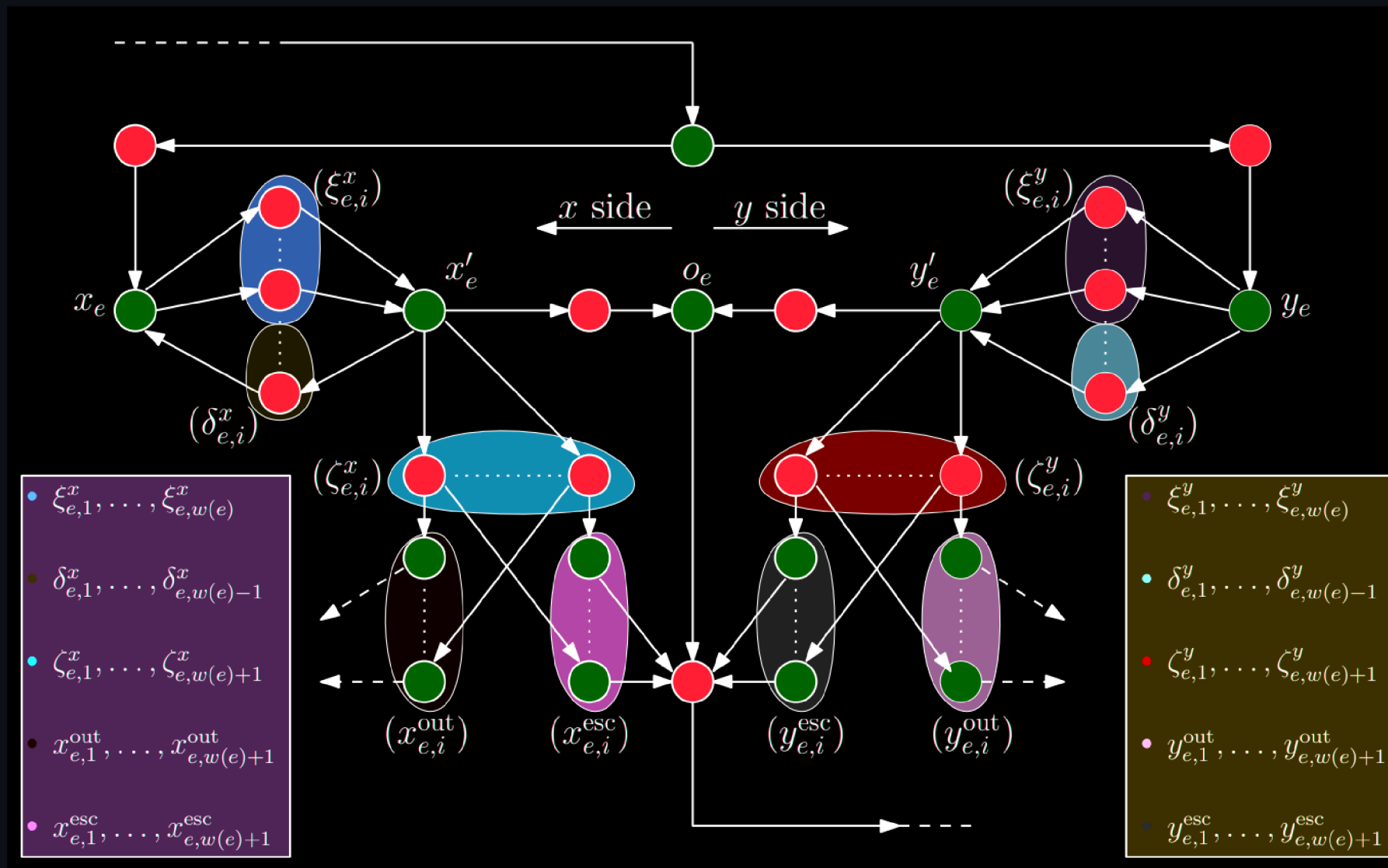
$$\text{choose the } y\text{-side} \iff x \rightarrow y$$

Phase II — Player 2 challenges a vertex

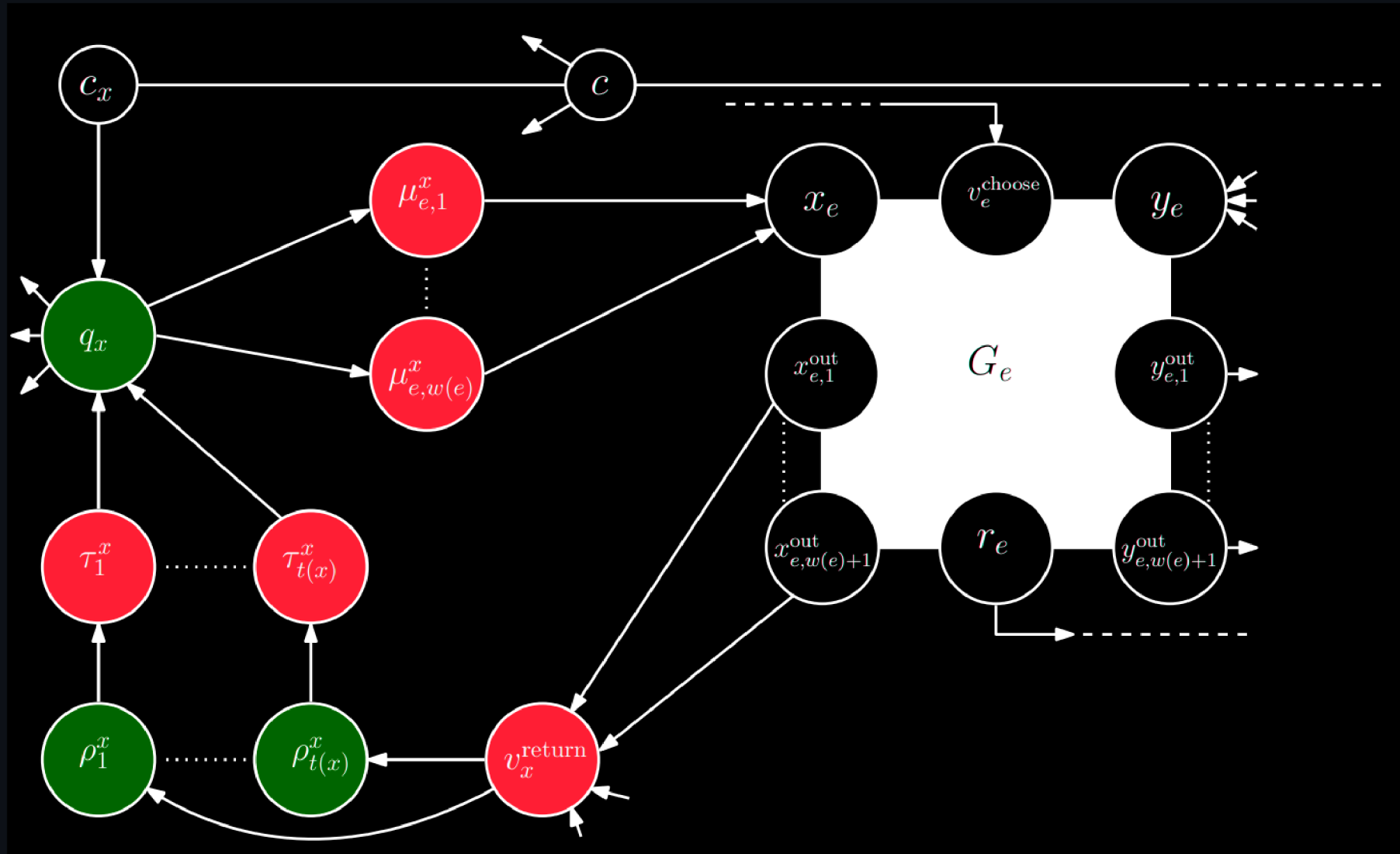
Player 2 chooses x and tests the total weight oriented out of x .

If outgoing weight $> t(x)$, Player 2 has one excursion too many.

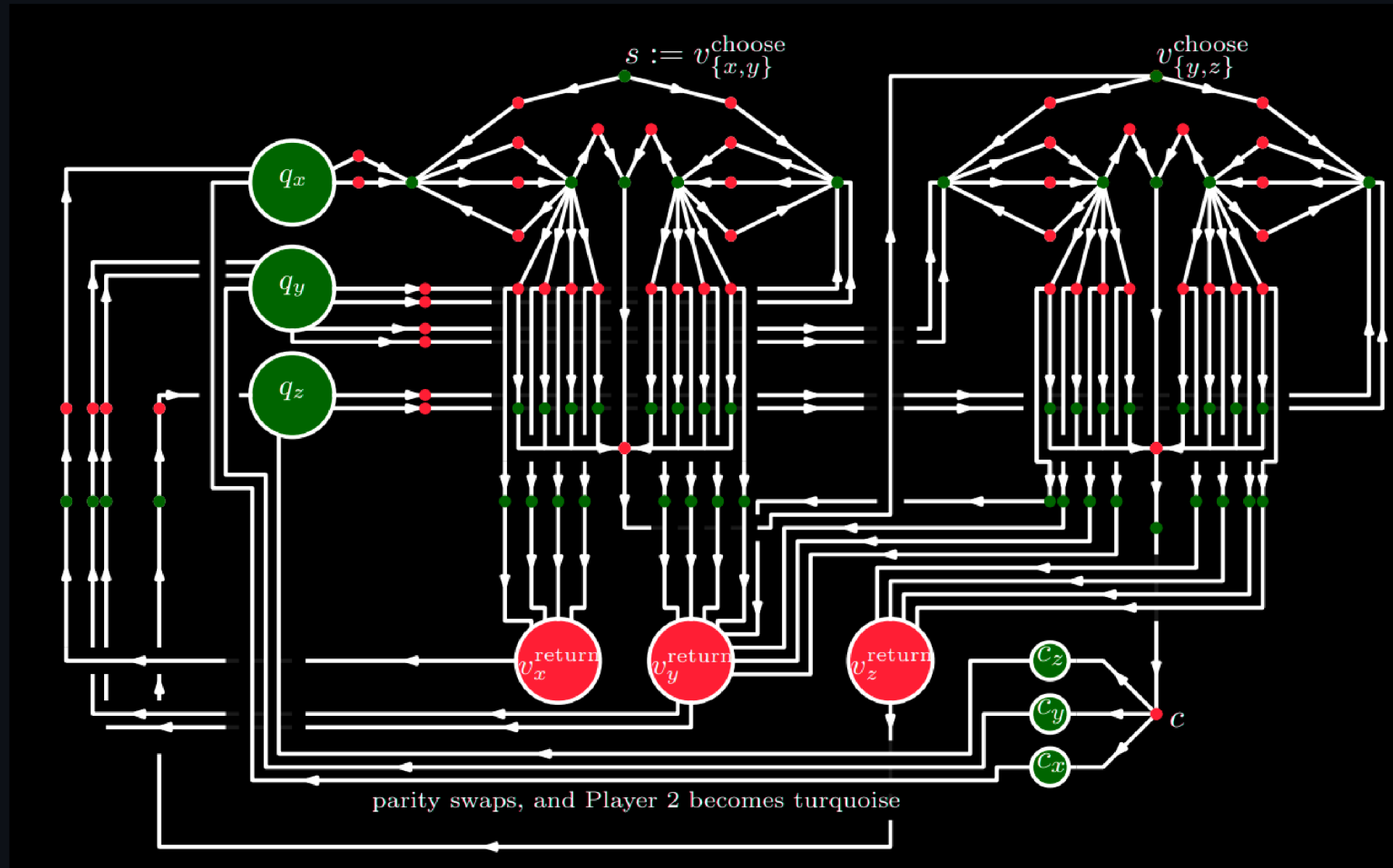
Phase 1



Phase 2



The full reduction for a small source instance



Why correctness follows

Feasible orientation

$$\sum_{e \in \theta_\omega(x)} w(e) \leq t(x) \text{ for every } x.$$

Player 2 eventually runs out of non-losing challenges.

Player 1 wins

Infeasible orientation

For some x ,

$$\sum_{e \in \theta_\omega(x)} w(e) > t(x).$$

Player 2 chooses this x and survives one challenge too many.

Player 2 wins

Why pathwidth is preserved

Process an input edge $e = \{x, y\}$ when x and y coexist in the source path decomposition.

Keep only the necessary anchors alive.

Introduce auxiliary gadget vertices locally, then forget them immediately.

$$\text{pw}(G') = O(k)$$

Directed to undirected

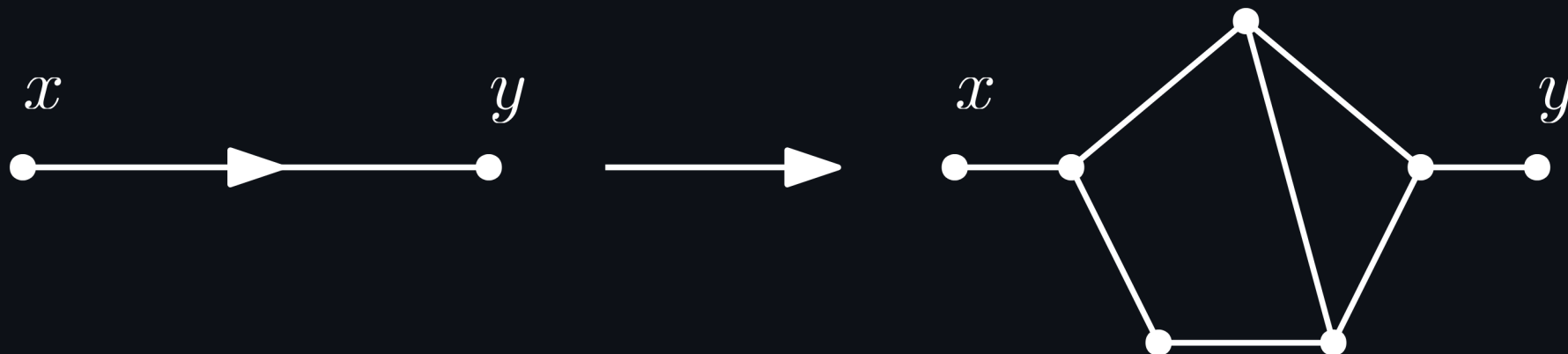
Each directed edge is replaced by a constant-size pseudoarc.

$$u \rightarrow v \quad \rightsquigarrow \quad \text{undirected pseudoarc}$$

Safe traversal works from u to v .

Entering the wrong way is losing under optimal play.

Directed hardness transfers to Undirected Edge Geography.



XP membership for width of treepartition

A rooted tree partition groups vertices into bags.

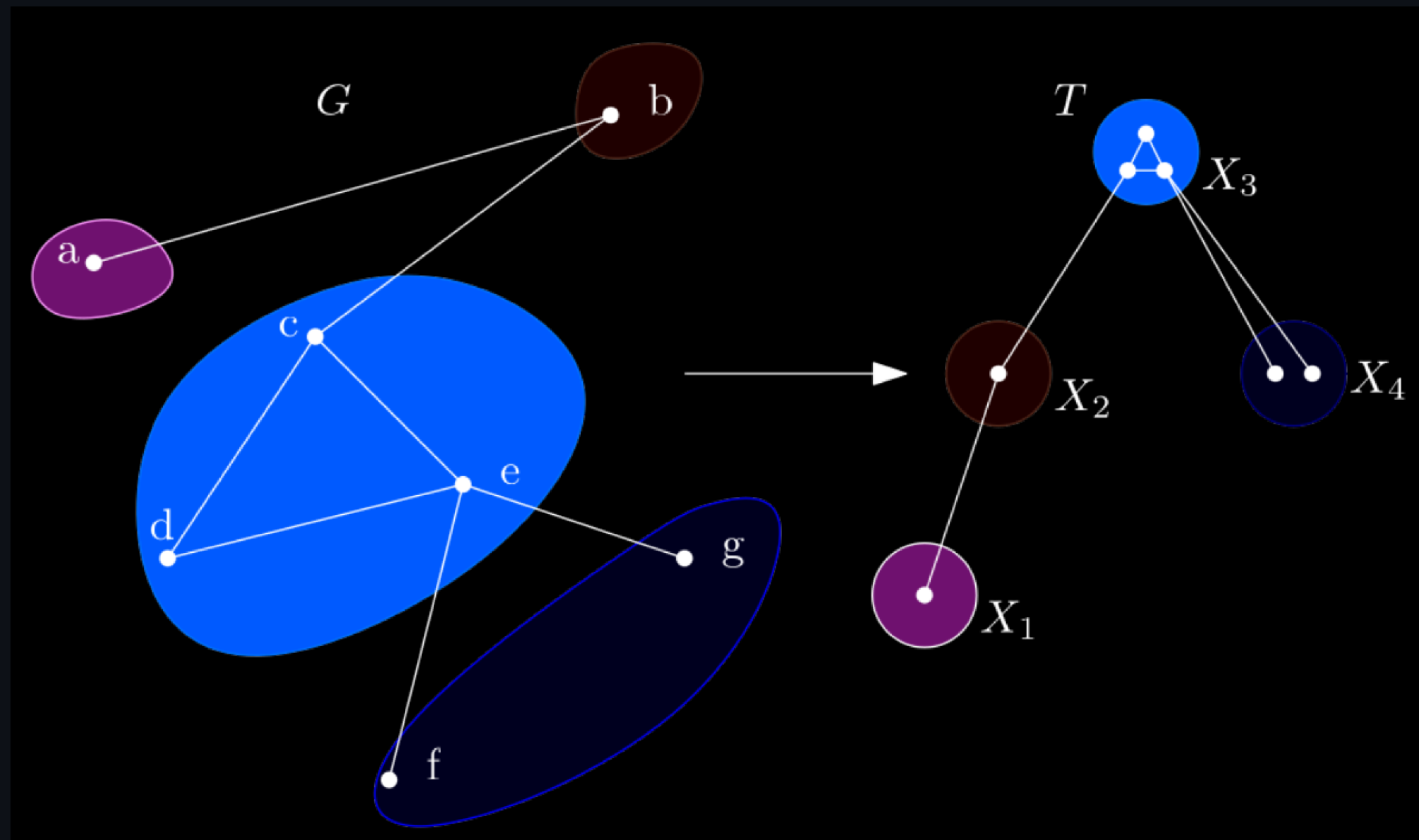
Edges are either:

- inside one bag, or
- between adjacent bags of the tree.

For width k , a parent-child cut has at most k^2 ports (on simple graphs).

Theorem

Undirected Edge Geography on simple graphs, given together with a rooted tree partition of width k , is solvable in XP time parameterized by k .



Bottom-up symbolic minimax

leaf $\longrightarrow$ interface type $\longrightarrow$ parent $\longrightarrow \dots \longrightarrow$ root

1 · Leaves

For every entry port and local residual state, brute-force minimax **inside the bag**.

Treat possible returns to the parent as variables ν_p .

2 · Internal bags

Replace each solved child by its interface type σ .

Collect equal types into counters m_σ and run local minimax.

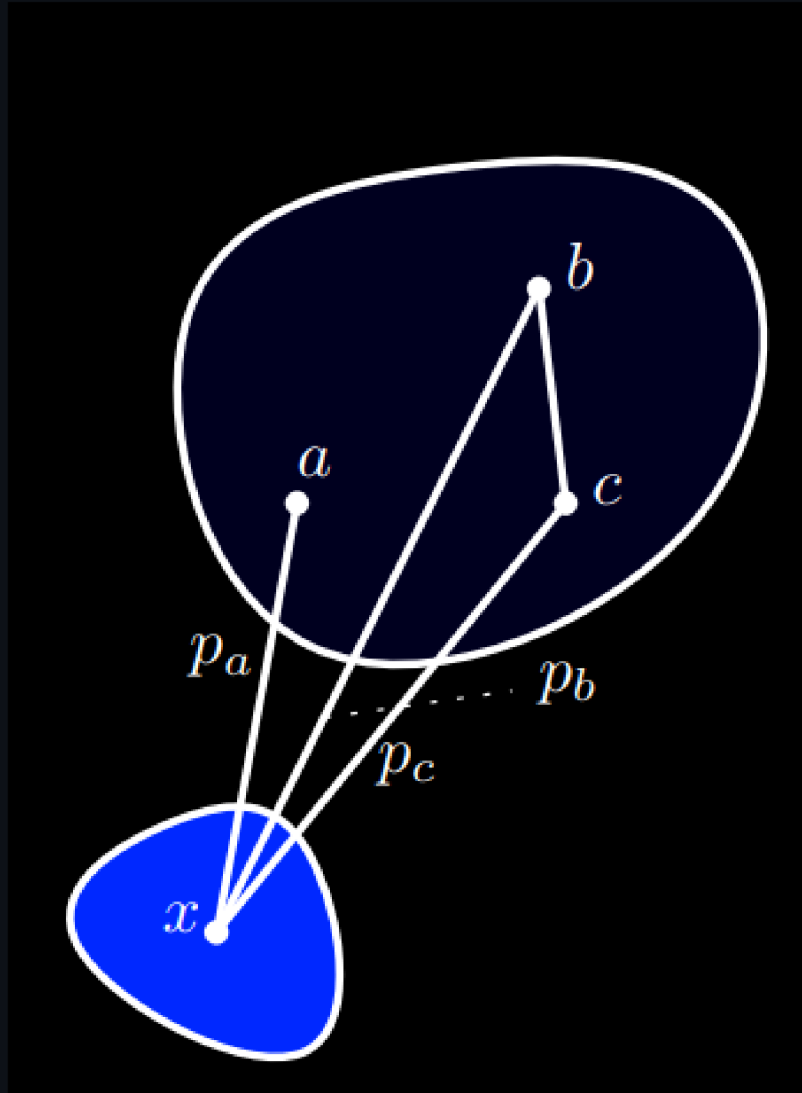
3 · Root

There is no unknown continuation above the root.

Evaluate the final compressed state to obtain the winner.

Entering a solved child is a **table lookup**; returning is a **counter update**.

A solved child conveys a boolean function



The child exports a response

$$\Phi_a(\nu_b, \nu_c) = \nu_b \wedge \nu_c.$$

ν_b and ν_c are the values of the two possible continuations in the parent.

The cut has $O(k^2)$ ports, so the truth table of Φ_a has size only $f(k)$.

Returning through p_b also exports the smaller residual type σ_c .

Child counters comprise the worst factor

The n -dependent part

For fixed k , there are t_k possible child types.

Each counter lies in $\{0, \dots, n\}$, so

$$m \in \{0, \dots, n\}^{t_k} \implies \#m \leq (n+1)^{t_k} = n^{O(t_k)}.$$

The other factors

Unused ports and internal edges contribute

$$2^{k^2} \cdot 2^{\binom{k}{2}} = 2^{O(k^2)}.$$

All other boundary data depends only on k ; collect it into $g(k)$.

$$\underbrace{n}_{\text{bags}} \cdot \underbrace{g(k)}_{\text{includes } 2^{O(k^2)}} \cdot \underbrace{n^{O(t_k)}}_{\text{counter states}} = n^{f(k)}.$$

Laughably bad in practice

Let T_r be the number of ambient types on an interface with r ports.

ports r	0	1	2	3	4
ambient types T_r	1	2	676	$\approx 2^{768}$	$\geq 2^{2^{4058}}$

Illustrative ambient count obtained by unwinding Definition 35.

The number of realizable types may be vastly smaller.

A width- k cut may have k^2 ports.

The crude bound is a tower of exponentials of height $O(k^2)$.

Takeaway

Pathwidth

small separators

**but residual history is
uncontrolled**

XNLP-hard

Tree-partition width

bounded interfaces

residual play can be compressed

XP

Open: Is Edge Geography in XP parameterized by pathwidth alone?